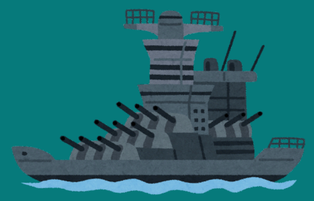
Battleship Game



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11:11PM

Introduction

Player vs Computer gameplay

No

Start

Battleship is a classic strategy game.

Fires shots to sink all enemy ships.

Hit shown as X, Miss shown as O

8x8 grid.

Agenda

Topics Covered

- To understand 2D arrays in C
- To use functions and modular programming
- To implement random number generation
- To apply loops and conditional statements
- To develop logical thinking through a game

Use of 2D Matrix in Project:

Game board represented using 2D array

Char board [8][8]

~ for water
S for ship
X for hit
O for miss

Separate matrices for player and computer

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```c
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

#define SIZE 8
#define NUM_SHIPS 5

int ship_sizes[NUM_SHIPS] = {5,4,3,3,2};

void init(char b[SIZE][SIZE]) {
    for(int i=0;i<SIZE;i++)
        for(int j=0;j<SIZE;j++)
            b[i][j]='~';
}

void print(char b[SIZE][SIZE], int hide) {
    printf("  ");
    for(int i=1;i<=SIZE;i++) printf("%d ",i)
    printf("\n");
    for(int i=0;i<SIZE;i++){
        printf("%c ", 'A'+i);
        for(int j=0;j<SIZE;j++){
            char ch=b[i][j];
            if(hide && ch=='S') ch='~';
            printf("%c ",ch);
```

```c
        }
        printf("\n");
    }
}

int canPlace(char b[SIZE][SIZE],int r,int c,int len,int dir){
    if(dir==0 && c+len>SIZE) return 0;
    if(dir==1 && r+len>SIZE) return 0;
    for(int i=0;i<len;i++)
        if(b[r+(dir?i:0)][c+(dir?0:i)]!='~') return 0;
    return 1;
}

void placeShip(char b[SIZE][SIZE],int len){
    int r,c,d;
    do{
        r=rand()%SIZE; c=rand()%SIZE; d=rand()%2;
    }while(!canPlace(b,r,c,len,d));
    for(int i=0;i<len;i++)
        b[r+(d?i:0)][c+(d?0:i)]='S';
}

void placeAllShips(char b[SIZE][SIZE]){
    for(int i=0;i<NUM_SHIPS;i++)
        placeShip(b,ship_sizes[i]);
}
```

```c
int fire(char b[SIZE][SIZE],int r,int c){
    if(b[r][c]=='S'){b[r][c]='X';return 1;}
    else if(b[r][c]=='~'){b[r][c]='O';return 0;}
    return -1;
}

int remaining(char b[SIZE][SIZE]){
    int cnt=0;
    for(int i=0;i<SIZE;i++)
        for(int j=0;j<SIZE;j++)
            if(b[i][j]=='S') cnt++;
    return cnt;
}

int main(){
    srand(time(0));
    char player[SIZE][SIZE], comp[SIZE][SIZE];
    init(player); init(comp);
    placeAllShips(player);
    placeAllShips(comp);

    char row; int col,pr,pc,cr,cc,res;
    printf("=== Battleship ===\n");
    while(1){
        printf("\nYour Board:\n"); print(player,0);
        printf("\nEnemy Board:\n"); print(comp,1);
```

```c
        // player turn
        printf("\nEnter target (e.g. B4): ");
        scanf(" %c%d",&row,&col);
        pr=row-'A'; pc=col-1;
        if(pr<0||pr>=SIZE||pc<0||pc>=SIZE){printf("Invalid!\n");continue;}
        res=fire(comp,pr,pc);
        if(res==1) printf("You hit!\n");
        else if(res==0) printf("You missed!\n");
        else {printf("Already fired!\n");continue;}

        if(remaining(comp)==0){printf("\nYou win!\n");break;}

        // computer turn
        do{cr=rand()%SIZE;cc=rand()%SIZE;}while(player[cr][cc]=='X'||player[cr][cc]=='O');
        res=fire(player,cr,cc);
        printf("Computer fires at %c%d: %s\n",'A'+cr,cc+1,res==1?"Hit!":"Miss!");

        if(remaining(player)==0){printf("\nComputer wins!\n");break;}
    }
    printf("\nFinal Boards:\nYour Board:\n");print(player,0);
    printf("\nEnemy Board:\n");print(comp,0);
    return 0;
}
```

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Future Scope

MENU

- Smarter computer firing strategy instead of random moves
- Create GUI version using graphics library
- Add sound effects and animations

FIREEEE!!!!

Conclusion

- Successfully implemented Battleship game in C
- Used arrays, functions, loops, and logic
- Improved understanding of matrix operations

- Practical application of C programming concepts

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Thank you!